

Two-year randomized, placebo-controlled study of black currant anthocyanins on visual field in glaucoma.

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AIM: To examine the influence of the black currant anthocyanins (BCACs) on the disease progression of open-angle glaucoma (OAG), a randomized, placebo-controlled, double-masked trial was made in 38 patients with OAG treated by antiglaucoma drops. METHODS: BCACs (50 mg/day, n = 19) or their placebos (n = 19) were orally administered once daily for a 24-month period. Systemic blood pressure, pulse rates, intraocular pressure (IOP), ocular blood circulation by laser-speckle flowgraphy, and Humphrey visual field mean deviation (MD) were measured during the 24-month period. RESULTS: As a main outcome measurement, we evaluated the difference between the groups in MD deterioration in the eye with a better MD from the trial's baseline through 24 months. A statistically significant difference was observed between the treatment groups in mean change from baseline in MD 24 months after therapy ($p = 0.039$, unpaired t test). Upon administration of BCACs, the ocular blood flows during the 24-month observational period increased in comparison with placebo-treated patients. However, no significant changes were observed in systemic and ocular conditions including IOP during the 24-month period. CONCLUSIONS: Our results suggest that oral administration of BCACs may be a safe and promising supplement for patients with OAG in addition to antiglaucoma medication. Copyright © 2012 S. Karger AG, Basel.